NewTen: Friction-Aware Tendon Transmission for GPU-Accelerated Dexterous Hand Simulation

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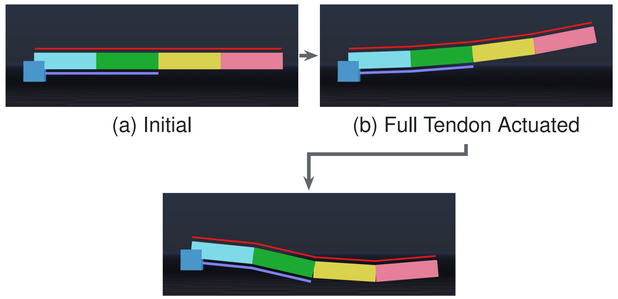
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Why Model Tendon Friction?

- **Idealized frictionless tendon models** neglect friction-induced tension losses and transmission hysteresis.
- **Detailed physical models can be computationally expensive**, particularly for large-scale robot learning requiring repeated simulation rollouts.
- **NewTen introduces a compact, stateful tendon-transmission model** coupled to Newton for GPU-accelerated hand simulation.

Tendon Routing and Actuation



(a) Initial (b) Full Tendon Actuated

(c) Full + Partial Actuation

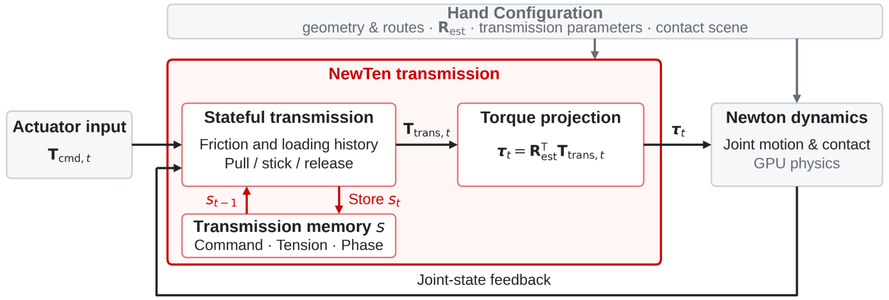
Full-Length Tendon Partial-Length Tendon

Commanded Tensions (Full / Partial):
(a) $0 / 0\,\mathrm{N}$, (b) $18 / 0\,\mathrm{N}$, (c) $18 / 58\,\mathrm{N}$.

- **Route geometry:** The full-length tendon spans all four links, while the partial-length tendon primarily actuates the proximal joints.
- **Combined actuation:** Engaging the partial-length tendon changes the bending profile, producing opposing bending directions along the articulated chain.

Simulation Framework

- **A shared configuration defines the hand model**, tendon routes, moment arms, and transmission parameters.
- **NewTen implements stateful, friction-aware tendon transmission**, mapping actuator commands and loading history to modeled transmitted tension and joint torques.
- **Newton applies the resulting joint torques** to simulate rigid-body motion and contact dynamics.



Results: Simulation

- **NewTen is evaluated using the ORCA dexterous hand** under identical actuator commands with tendon-friction coefficients $\mu = 0$ and $\mu = 0.25$.

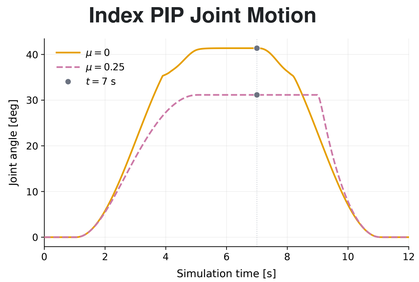


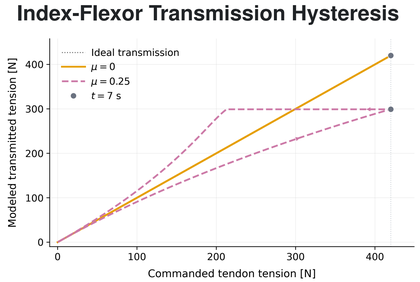
$\mu = 0$ $t = 7$ s $\mu = 0.25$ $t = 7$ s

- **At $t = 7\,\mathrm{s}$, the friction-aware simulation exhibits reduced index PIP flexion** under identical actuator commands.

Results: Quantitative Comparison

- The **friction-aware model ($\mu = 0.25$) exhibits transmission hysteresis**, producing different transmitted tensions during loading and unloading at the same commanded tension.





- At $t = 7\,\mathrm{s}$, with a $420\,\mathrm{N}$ index-flexor command, $\mu = 0.25$ produces **28.8% lower modeled transmitted tension and 10.22° less index PIP flexion** than $\mu = 0$.

Values at $t = 7\,\mathrm{s}$

Quantity	$\mu = 0$	$\mu = 0.25$
Index PIP angle [deg]	41.37	31.15
Modeled transmitted tension [N]	420.00	299.04

- The simulation comparison uses constant estimated moment arms and uncalibrated transmission parameters.

Summary

- **NewTen provides a reusable, stateful tendon-transmission framework** for modeling friction-dependent actuation in Newton-based robotic hand simulation.
- **ORCA simulations demonstrate model-generated transmission hysteresis** and friction-dependent changes in joint motion under identical actuator commands.

Future Development

- **Collect physical hand-motion data** and calibrate NewTen and a frictionless baseline independently using the same calibration dataset.
- **Compare prediction accuracy on held-out motions** and quantify NewTen's computational overhead relative to the baseline.